Where the baseline fails

Walk-forward predictions, December 2025 to May 2026. Both tracks contain the same **10,069 validation homes**; 20,138 prediction rows do not represent independent homes.

Track	MdAPE	MAPE	Median signed	90th pct. APE
Off-market	20.84%	27.43%	+8.07%	57.99%
List-aware	7.26%	8.09%	+5.68%	15.89%

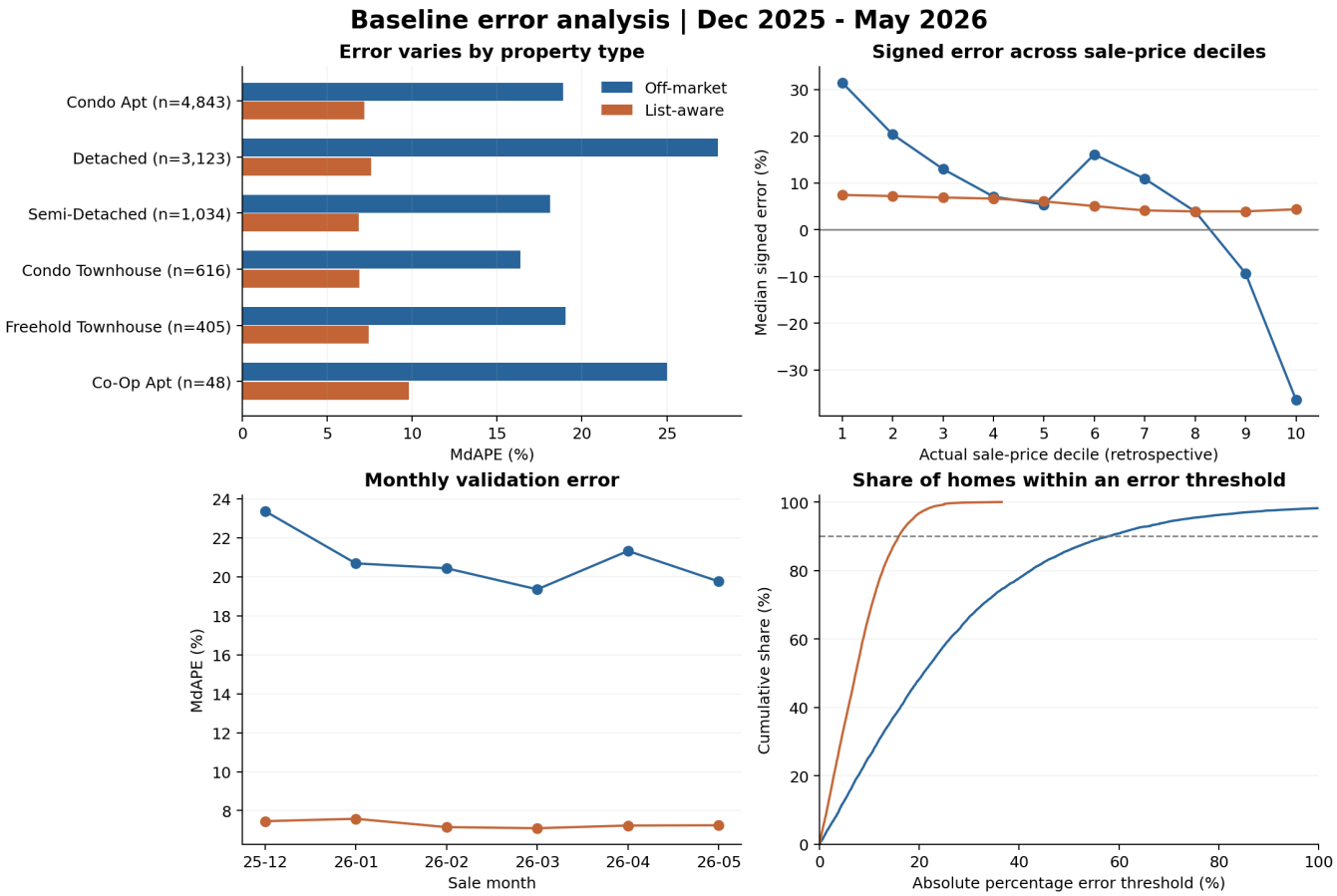


Figure 1. Positive signed error means overprediction. Sale-price deciles are retrospective groups defined using the outcome. The cumulative-error panel is shown up to 100% APE.

Detached homes deserve priority. Off-market MdAPE is 27.99% for detached homes (n=3,123), versus 16.39% for condo townhouses (n=616). Co-op results are unstable with only 48 homes.

Bias differs across prices. The cheapest and dearest sale-price deciles have median signed errors of +31.44% and -36.40%. Conditioning on the outcome can contribute to this pattern; it is not causal evidence. List-aware overpredicts 76.03% of homes.

Coverage can hide weak groups

Nominal **90% prediction intervals** were evaluated on 8,045 eligible homes per track. The other 2,024 homes lacked enough historical residuals: 1,238 December warmup homes plus 786 with insufficient history at listing time.

Measure	Off-market	List-aware
Observed coverage	90.11%	89.36%
Median width / point prediction	114.23%	30.04%
Median width in dollars	$1,067,401	$259,862

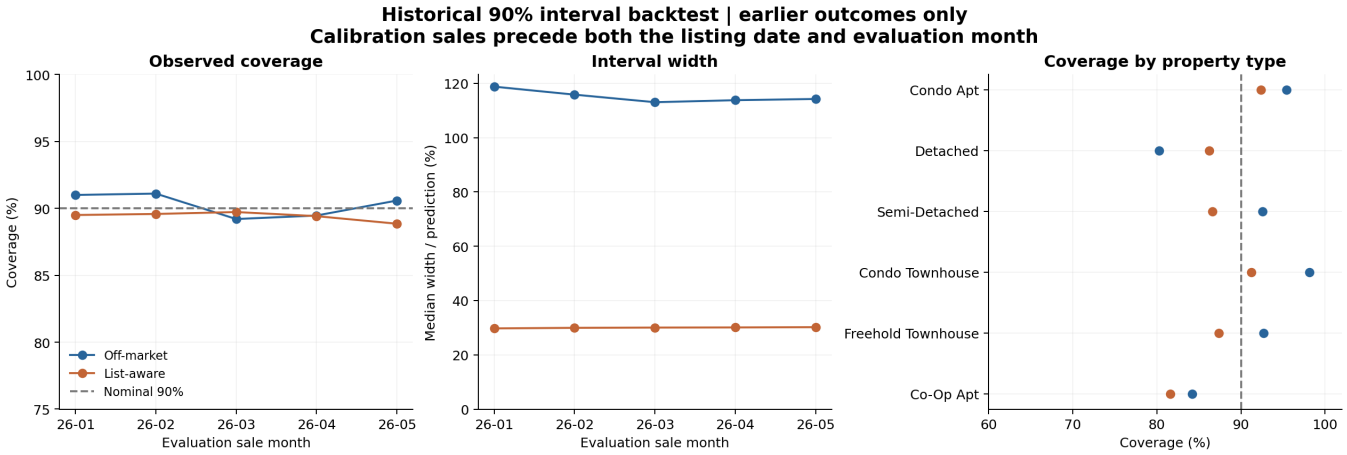


Figure 2. These intervals preserve the baseline point predictions. Width is upper minus lower, divided by the point prediction. Coverage is conditional on calibration eligibility.

A close overall result is not enough

Property type	Evaluated n	Off-market	List-aware
Detached	2,652	80.28%	86.20%
Condo Apt	3,621	95.44%	92.38%
Condo Townhouse	480	98.13%	91.25%

Construction. For each track, use s = |log(sold / predicted)| from sales strictly before both the target listing date and the evaluation month. Require n >= 100; select the k-th ordered score, k = ceil((n+1) x 0.90), then return [prediction x exp(-q), prediction x exp(q)].

Interpretation. This historical, conformal-style interval is an empirical backtest, not a strict future-coverage guarantee. Time drift and changing fitted models can violate exchangeability. Wide off-market intervals and low detached coverage motivate testing property-aware calibration on later data.

Evidence and next steps

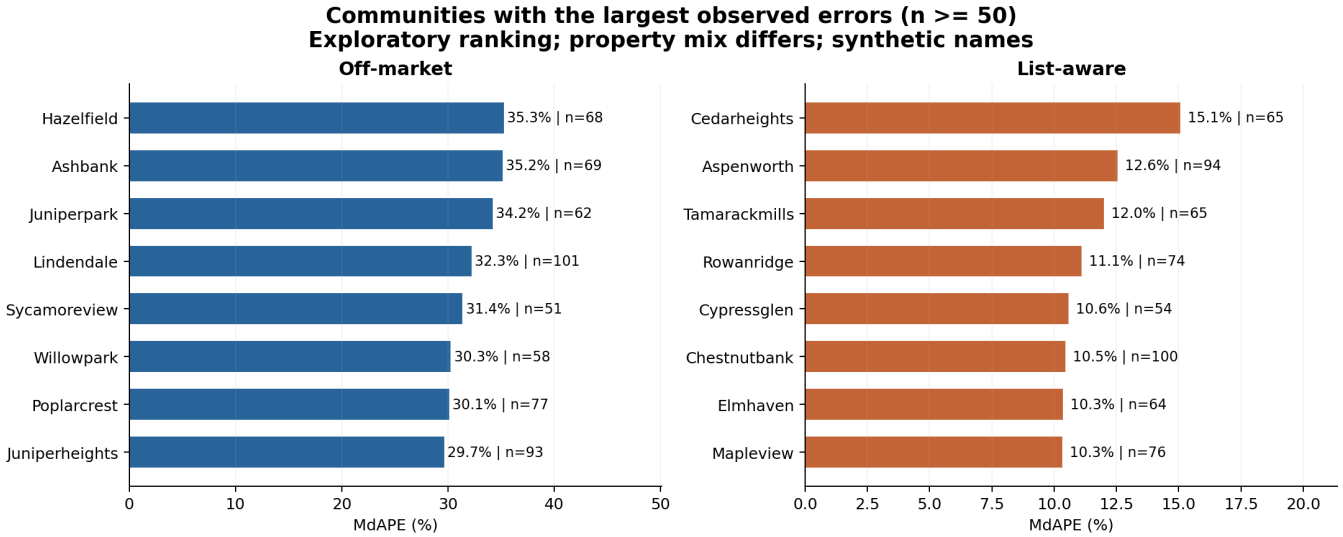


Figure 3. Only communities with at least 50 validation homes are ranked. These exploratory differences are unadjusted for property mix, month and price; names and locations are synthetic.

A reconstructed failure

SYN-0092314: sold for $844,000; predicted $4,317,900; APE 411.60%. Its area was 1,166 sq ft. Rebuilding the official ten-nearest-sales prediction gave the same result. The comparables had a median area of 1,883.5 sq ft; only 3 of 10 shared the community, and sales ranged from March 2021 to September 2025. Nearby, same-type homes can still be poor comparables. This does not isolate the causal effect of any one mismatch.

Distinguish score uncertainty from price uncertainty

95% MdAPE bootstrap range	Off-market	List-aware
Individual-home resampling	20.41-21.36%	7.12-7.42%
Whole-month resampling	19.84-21.93%	7.16-7.40%

1,000 resamples; seed 42. Six month clusters provide only a rough sensitivity check. These ranges describe the aggregate metric, not uncertainty for an individual house.

Actions for the final model

1. Test size, community and recency adjustments for off-market comparables. **2.** Test list-price bias correction inside training folds. **3.** Evaluate property-aware intervals with chronological calibration. None of these improvements has been demonstrated here; rerun the diagnostics on the final model before making performance claims.

Provenance and limits. Supplied baseline predictions matched the reported scores. All list-aware predictions equal list price; 24 sampled off-market rows plus 3 extreme cases were reconstructed successfully. Full training provenance was not audited. No hidden-test outcomes or launch-video content were used.

Sources: supplied baseline_results.zip and case handout; official synthetic listings; starter repository; Angelopoulos & Bates, conformal prediction introduction. Full tables, input hashes and executable analysis accompany this brief.